

claude-windows / fa4dc4eb-4705-47fd-b044-4f017b586b15

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\fa4dc4eb-4705-47fd-b044-4f017b586b15\subagents\workflows\wf_ac0cbb29-6bf\agent-  
aee626016177f5958.jsonl`  
- SHA-256: `331b0ebf94455cdfbd9681e17848759c45795f52a67ba6ca67551a2a4da2e679`  
- Source modified: `2026-06-19T15:49:15+00:00`  
- Imported at: `2026-07-05T16:48:22+00:00`  
- project: `wf_ac0cbb29-6bf`  
- session_id: `fa4dc4eb-4705-47fd-b044-4f017b586b15`
```

Transcript

1. user (2026-06-19T15:46:34.350Z)

You are auditing a badminton rally-detection codebase for DOC/COMMENT STALENESS ? statements that CONTRADICT the current code or data.

HARD RULES:

- Report ONLY factual staleness with CODE EVIDENCE (file:line or symbol) that proves it. Open the cited code and confirm before reporting.
- NEVER invent a fact or a status. If a thing is built-but-default-OFF / unwired / unmeasured, say EXACTLY that ? do not overclaim "in production".
- A "[design]" tag is stale ONLY if the thing is actually BUILT in code (cite the symbol).
- DO NOT propose subjective rewrites, style/tone changes, or improvements. DO NOT relitigate owner-gated DECISIONS (owned-model roadmap, licensing/MIT-Apache stance, product/strategy choices, genuinely-still-design proposals).
- Preserve the document's voice and formatting; make the SMALLEST surgical edit. stale_text MUST be copied verbatim with enough context to be unique.
- IGNORE docs/archives/** (terminal snapshots) and any memory/ file.

Working dir is the repo root; use Read/Grep/Glob freely.

YOUR SLICE ? meta:

Top-level meta staleness vs code: CLAUDE.md, docs/README.md, docs/NEXT_STEPS.md. SEED 1 (verify): CLAUDE.md says "This dev container lacks GPU / torch / cv2 / numpy ? the full pipeline can not run here" ? STALE: the repo runs the full WSL/GPU/WASB pipeline and cv2/numpy/torch import fine here. SEED 2 (verify against backend/config/models.py): the cap=6 + R4 inactivity-end milestone is now DEFAULT-ON (max_window_duration=6.0, window_inactivity_end=1.5, inactivity_rally_thresh=0.20, inactivity_min_density=0.30) ? flag any meta doc still calling local windowing not-started / default-OFF / not-flipped / scaffolding-only. Only fix factual contradictions, not roadmap intent.

Return EVERY code-contradicting staleness item you can verify in your slice. Exhaustive within the slice; facts only.

2. assistant (2026-06-19T15:46:37.047Z)

I'll audit the three meta docs in my slice for staleness against the current code. Let me start by reading the key files and verifying the seeds.

3. user (2026-06-19T15:46:38.062Z)

```
1      """Pydantic models for the application configuration.  
2  
3      This module defines the canonical, validated, typed configuration for the  
4      Badminton Highlight Indexer. Built-in defaults live here (single source of truth).  
5  
6      The models support graceful migration from old config.json files via extra="allow".  
7      """  
8
```

```

9     from __future__ import annotations
10
11     from typing import Any, List, Literal, Optional
12
13     from pydantic import BaseModel, Field, field_validator, model_validator
14
15     Sport = Literal["badminton", "tennis", "pickleball", "table_tennis", "cricket"]
16
17
18     class ValidationRelevanceConfig(BaseModel):
19         court_green_lower: list[int] = Field(default=[35, 40, 40])
20         court_green_upper: list[int] = Field(default=[85, 255, 255])
21         court_pixels_min_ratio: float = Field(default=0.05, ge=0.0, le=1.0)
22
23
24     class ValidationConfig(BaseModel):
25         min_resolution_width: int = 1280
26         min_resolution_height: int = 720
27         min_fps: float = 29.9
28         min_blur_threshold: float = 20.0
29         max_scene_cuts_per_minute: float = 0.5
30         relevance: ValidationRelevanceConfig =
31         Field(default_factory=ValidationRelevanceConfig)
32
33     class TrackNetConfig(BaseModel):
34         wsl_distro: str = "Ubuntu"
35         repo_dir: str = "~/models/TrackNetV4"
36         conda_sh: str = "~/miniconda3/etc/profile.d/conda.sh"
37         conda_env: str = "TrackNetV4"
38         weights_path: str = ""
39         queue_length: int = 5
40         stage_in_wsl: bool = True
41         wsl_stage_dir: str = "~/clips"
42         timeout_sec: int = 1800 # 30 min; kills a hung WSL/GPU call (0 = no timeout)
43         velocity_threshold: float = 0.02
44         # Cache hygiene: after a SUCCESSFUL run the staged video copy (and, for WASB, the
45         # decoded frame cache) are deleted automatically ? they are pure intermediates,
46         # regenerable from the source, and the dominant disk consumer (~GBs/video). Set
47         # true only for debugging. On FAILURE they are always kept so a re-run can resume.
48         keep_frames: bool = False
49         # Warn before a run if WSL free space (at wsl_stage_dir) is below this many GB.
50         # 0 = disabled.
51         min_free_gb: float = 0.0
52
53
54     class WasbIndexConfig(BaseModel):
55         """Typed config for the live WASB shuttle substrate (indexing.wasb).
56
57         Mirrors backend/pipeline/detectors/wasb_runner.py::WasbConfig.from_indexing_cfg so
58         these knobs actually take effect ? previously this whole block was silently dropped
59         because nested config sections defaulted to extra='ignore'.
60         """
61         wsl_distro: str = "Ubuntu"
62         repo_dir: str = "~/models/WASB-SBDT"
63         conda_sh: str = "~/miniconda3/etc/profile.d/conda.sh"
64         conda_env: str = "wasb"
65         weights_path: str = "~/models/WASB-
66         SBDT/pretrained_weights/wasb_badminton_best.pth.tar"
67         sport: str = "badminton"
68         wsl_stage_dir: str = "~/clips"
69         timeout_sec: int = 1800 # 30 min; kills a hung WSL/GPU call (0 = no timeout)
70         velocity_threshold: float = 0.02
71         # Cache hygiene: after a SUCCESSFUL run the staged video copy + decoded frame cache
72         # (the ~GBs/video disk hog) are deleted automatically ? pure intermediates,

```

```

regenerable
72     # from the source. Set true only for debugging. On FAILURE they are kept so a re-run
resumes.
73     keep_frames: bool = False
74     # Warn before a run if WSL free space (at wsl_stage_dir) is below this many GB. 0 =
disabled.
75     min_free_gb: float = 0.0
76     # FAST PATH (default OFF until owner-verified side-by-side): decode the staged video
77     # on the fly and feed frames straight to the detector, skipping the slow per-frame
PNG
78     # extraction that dominates a long clip (~46k PNGs for a 12-min 1080p60 video, which
79     # overran the 30-min timeout). Output is bit-identical to the PNG path (PNG is
lossless),
80     # so this is purely a speed/disk win; kept opt-in until the side-by-side confirms
it.
81     stream_video: bool = False
82
83
84     class FusionConfig(BaseModel):
85         """Config for the multi-signal `fusion_hybrid` segmenter (MULTI_SIGNAL_FUSION_PLAN
P2).
86
87         Every default reproduces control behaviour: the fusion segmenter persists the
88         shuttle net-crossing count (Gate A signal) and ? only when ``cue_flags['person']``
is
89         true ? the person-cue features, but it does NOT gate the served handoff unless a
90         threshold is raised (``min_crossings``/``min_players`` default 0 = OFF). The court
mask
91         is optional: ``court_polygon=None`` ? Gate B counts every detection (player-count-
only);
92         supplied ? off-court persons (spectators / adjacent courts) are excluded.
93         """
94         # Which trajectory substrate seeds the windows (shuttle stays primary).
95         substrate: Literal["wasb", "tracknet"] = "wasb"
96         # Windowing velocity threshold for the fusion family (the engine reads
97         # indexing.<family>.velocity_threshold). Default matches the substrates' 0.02; set
98         # equal to indexing.wasb.velocity_threshold to A/B fusion against a tuned wasb run.
99         velocity_threshold: float = 0.02
100        # Per-cue enable flags, e.g. {"person": true}. Person cue is OFF unless explicitly
101        # enabled ? so the default path never imports torch / touches the GPU.
102        cue_flags: dict[str, bool] = Field(default_factory=dict)
103        # Served Gate A: drop windows whose shuttle net-crossings < this from the AI
handoff.
104        # 0 = OFF (no gating; persisted candidates are always the full superset for offline
re-scoring).
105        min_crossings: int = Field(default=0, ge=0)
106        # Served Gate B: when the person cue is on, drop windows with median in-court
players
107        # < this. 0 = OFF.
108        min_players: int = Field(default=0, ge=0)
109        # Optional court mask as normalised 0-1 [[x, y], ...] polygon. None = no mask.
110        court_polygon: Optional[list[list[float]]] = None
111        # Net line for the person two-sided-motion split (person features only ? the shuttle
112        # net-crossing gate keeps rally_gate's camera-agnostic auto-estimate).
113        net_axis: Literal["x", "y"] = "y"
114        net_position: float = Field(default=0.5, ge=0.0, le=1.0)
115
116
117        class IndexingConfig(BaseModel):
118            default_segmenter: str = "yolo_hybrid"
119            use_gpu: bool = True
120            motion_threshold: float = 0.08
121            min_segment_duration: float = 3.0
122            dead_time_merge_gap: float = 2.0
123            chunk_duration: float = 90.0

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124     chunk_overlap: float = 10.0
125     detector_model: str = "fasterrcnn_resnet50_fpn"
126     detector_score_threshold: float = 0.5
127     yolo_velocity_threshold: float = 0.15
128     yolo_frame_skip: int = 3
129     yolo_tracker: str = "bytetrack.yaml"
130     yolo_prefetch_workers: int = 2
131     min_action_window_duration: float = 2.0
132     # BOUNDED WINDOWING (over-merge fix W1, RALLY_QUALITY_RESEARCH.md §6). 0 = OFF; > 0
= a window
133     # longer than this many seconds is recursively split at its largest internal
inactivity gap (?
134     # ``window_min_split_gap`` s ? the most likely rally boundary), so one window can't
swallow
135     # several rallies + the dead time between them. Never merges, only cuts (recall
can't drop).
136     # *** R1 MILESTONE DEFAULT-ON (cap=6): the smallest safe local-windowing flip.
Measured n=13
137     # golden, R2 tolF1: baseline?milestone (cap=6 + R4 below) ?+0.222, sign 12/0,
p=0.0005; cap=6
138     # beats cap=12 ?+0.110, sign 12/0; net over-seg guard PASS (mean merge_rate
0.651?0.161). ***
139     max_window_duration: float = 6.0
140     window_min_split_gap: float = 0.5 # min internal gap (s) that qualifies as a split
point
141     # HARD-CAP fallback for W1: when True, an over-long window with NO internal
inactivity gap
142     # (a continuously-active trajectory ? e.g. the real-footage mahadevpura-1 174s blob)
is
143     # chopped at its midpoint until ? max_window_duration, making the cap a true upper
bound.
144     # Only cuts (recall can't drop). OFF by default until measured/promoted.
145     window_hard_cap: bool = False
146     # R4 rally-END inactivity trim (s, 0 = OFF). After a rally the shuttle keeps moving
slowly
147     # (picked up / walked) above velocity_thresh, so the window over-extends its END
(the measured
148     # [?end] gap vs golden). Trim the post-rally tail back to the last frame whose
trailing
149     # window stays dense with FAST motion (velocity > inactivity_rally_thresh). Only
cuts.
150     # *** R1 MILESTONE DEFAULT-ON (1.5/0.20/0.30): on top of cap=6, R4 adds a small but
significant
151     # end-precision gain ? ?tolF1 +0.040, sign 9/1/3, p=0.022, [?end] 4.96?3.31s (n=13).
The W1 cap
152     # is the dominant lever; R4 is the marginal increment. See QUALITY_ITERATIONS.md
"R1". ***
153     window_inactivity_end: float = 1.5
154     inactivity_rally_thresh: float = 0.20
155     inactivity_min_density: float = 0.30
156     # R5 blob-fallback (default OFF). LAST-RESORT splitter for the continuously-active
blob: after
157     # the W1 gap-split AND the R4 inactivity trim have each had their chance, a group
STILL longer
158     # than window_blob_factor × max_window_duration (no internal gap, no rally-end
signal ? e.g.
159     # mahadevpura-1's ~65s residual ? 11× a 6s cap) is midpoint-chopped to ? the cap as
a FORCED cut
160     # (logged + flagged on the window) so the degenerate single-window outcome is
avoided and the
161     # clip is surfaced as needing rally-STATE cues, not a bigger cap. Differs from
window_hard_cap
162     # (which chops BEFORE R4 and over-segments normal clips): blob-fallback runs AFTER
R4 and only
163     # force-cuts the genuine blob (the factor gate). Only cuts (recall can't drop).

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164     window_blob_fallback: bool = False
165     # Magnitude gate that keeps R5 surgical: only a group longer than this x
max_window_duration is a
166     # blob (a normal long rally ~1-2x the cap is left alone; the mahadevpura-1 residual
is ~11x).
167     # Default 4.0 is do-no-harm on the golden corpus (rescues only the continuously-
active
168     # clips, every other clip a no-op within noise) ? see docs/QUALITY_ITERATIONS.md
"R5".
169     window_blob_factor: float = 4.0
170     log_yolo_candidates: bool = True
171     store_yolo_candidates: bool = True
172     ai_handoff_padding: float = 2.0
173     ai_max_workers: int = 5
174     # Width (px) the gemini segmenter re-encodes chunks to before upload. Stream-copied
175     # chunks of a high-bitrate (4K) source blow past the provider upload cap
176     # (backend/providers/gemini.py::MAX_UPLOAD_MB) and are refused ? observed live as
177     # "0 segments / success". Re-encoding also makes chunk boundaries frame-accurate
178     # (stream copy cuts at the nearest keyframe). 0 = legacy stream-copy at source res.
179     # Matches gemini_refine's --clip-scale-width default.
180     ai_chunk_scale_width: int = 1280
181     # FULLY-LOCAL mode (default OFF): when true, the trajectory-hybrid segmenters
182     # (wasb_hybrid / tracknet_hybrid / fusion_hybrid) skip the Phase-3 AI handoff
entirely and
183     # emit their local candidate windows AS the rallies ? no provider, no API key,
nothing
184     # uploaded. Lets the local detector run standalone (e.g. to benchmark it against the
Gemini
185     # path, or to operate with zero cloud dependency). The pure `gemini` segmenter
ignores this.
186     skip_ai_handoff: bool = False
187     # Optional debug knob: trim every video to the first N seconds before processing.
188     debug_duration: Optional[float] = None
189
190     tracknet: TrackNetConfig = Field(default_factory=TrackNetConfig)
191     wasb: WasbIndexConfig = Field(default_factory=WasbIndexConfig)
192     fusion: FusionConfig = Field(default_factory=FusionConfig)
193
194     @field_validator("default_segmenter")
195     @classmethod
196     def segmenter_must_be_registered(cls, v: str) -> str:
197         # Registry check happens at runtime in main/segmenter selection.
198         # Here we just allow any string for forward compatibility.
199         return v
200
201     @model_validator(mode="after")
202     def _chunk_step_must_be_positive(self) -> "IndexingConfig":
203         # The chunked segmenters advance by (chunk_duration - chunk_overlap); a non-
positive
204         # step makes `while t < duration: t += step` loop forever, growing memory before
any
205         # GPU work. Reject the bad config at load time instead.
206         if self.chunk_overlap >= self.chunk_duration:
207             raise ValueError(
208                 f"indexing.chunk_overlap ({self.chunk_overlap}) must be < chunk_duration
"
209                 f"({self.chunk_duration}); otherwise chunking never advances."
210             )
211         return self
212
213
214     class OutputConfig(BaseModel):
215         default_highlights_name: str = "highlights.mp4"
216         db_name: str = "sports_indexer.db"
217         # Directory where the DB, highlight reels, and processed clips are written.

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218         # The CLI's --output-dir overrides this at startup (see backend/main.py::main).
219         output_dir: str = "output"
220
221
222     class WatermarkConfig(BaseModel):
223         """Branding overlay burned into each highlight clip during the per-clip re-encode
224         (backend/pipeline/stitcher/compiler.py). **On by default** ? set `enabled: false`
225         (under `stitching.watermark` in config.json) to turn it off. The text is fully
226         configurable. The concat stage stays stream-copy (-c copy)."""
227         enabled: bool = True
228         text: str = "Generated by Khelsutra.guru"
229         logo_path: str = ""          # optional transparent PNG overlay; "" = no logo
230         font_path: str = ""          # optional .ttf; "" = use the fontconfig `font` family
231         font: str = "Sans"           # fontconfig family used when font_path is empty
232         font_size_ratio: float = Field(default=0.035, gt=0.0, le=0.5) # text height as a
fraction of frame height
233         opacity: float = Field(default=0.85, ge=0.0, le=1.0)
234         box: bool = True              # faint backing box behind the text for legibility
235         margin: int = Field(default=24, ge=0) # px inset from the chosen corner
236         position: Literal["bottom-right", "bottom-left", "top-right", "top-left"] = "bottom-
right"
237
238
239     class StitchingConfig(BaseModel):
240         begin_padding: float = 0.5
241         end_padding: float = 0.5
242         use_cache: bool = False
243         min_shot_count: int = 1
244         # Long-rally reel filter (seconds; 0 = OFF). Drop detected rally pairs whose
duration
245         # (end ? start) is below this from the highlight reel, BEFORE compiling. A cleaner
246         # reel-selection knob than `min_shot_count`: shot_count is unlabeled / "Unknown" on
the
247         # fully-local no-AI path, whereas duration is derived from the rally's own
start/end. The
248         # local detector over-fires and its false positives are mostly SHORT (motion blips,
249         # background-court fragments), so this keeps the reel to the eye-catching long
rallies.
250         # OFF by default ? the detector output is unchanged, only which rallies reach the
reel.
251         min_rally_duration: float = Field(default=0.0, ge=0.0)
252         watermark: WatermarkConfig = Field(default_factory=WatermarkConfig)
253
254
255     class LoggingConfig(BaseModel):
256         """Logging knobs for the run entrypoints (see backend/utils/logging_setup.py)."""
257         # Third-party HTTP/RPC client loggers (httpx, httpcore, urllib3, google-genai,
258         # google.auth, grpc) emit one INFO line per request ? dozens per Gemini run, which
259         # buries our own [rally]/[compile] lines in run.log during manual QA. Default raises
260         # them to WARNING; set true to restore full chatter for request-flow debugging.
261         verbose_http: bool = False
262
263
264     class SecurityConfig(BaseModel):
265         # When set, /api/process video_path must resolve under this directory (hosted/tenant
266         # mode). Default None preserves the single-user local 'any local file' workflow.
267         allowed_video_root: Optional[str] = None
268
269         # --- Chunked upload (B2, PR-2) ---
270         # Landing zone for browser uploads (assembled from parts). Per-user subdirectories
271         # arrive with PR-3 identity scoping. ? In hosted mode, set allowed_video_root to
272         # contain upload_dir or /api/process will reject the uploaded paths.
273         upload_dir: str = "output/uploads"
274         # Whole-file cap (default 4 GB) and per-part cap. Parts stay under ~100 MB because

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275     # free-tier edge proxies cap request bodies there (HOSTING_PLAN $2).
276     max_upload_mb: int = 4096
277     max_part_mb: int = 95
278     allowed_upload_extensions: List[str] = ["mp4", "mov"]
279
280
281     class AppConfig(BaseModel):
282         """Root configuration object.
283
284         All runtime configuration should be accessed via an instance of this class
285         (or the loader) rather than raw dict lookups.
286         """
287
288         sport: Sport = "badminton"
289         ai_provider: str = "gemini"
290         # Auto-tracking alias: 2.5-flash is legacy now that 3.5-flash is the stable
flagship;
291         # the alias never deprecates underneath us. Eval reports record the run date
alongside.
292         ai_model: str = "gemini-flash-latest"
293
294         validation: ValidationConfig = Field(default_factory=ValidationConfig)
295         indexing: IndexingConfig = Field(default_factory=IndexingConfig)
296         output: OutputConfig = Field(default_factory=OutputConfig)
297         stitching: StitchingConfig = Field(default_factory=StitchingConfig)
298         security: SecurityConfig = Field(default_factory=SecurityConfig)
299         logging: LoggingConfig = Field(default_factory=LoggingConfig)
300
301         # Allow extra keys during transition so existing config.json files do not break.
302         model_config = {
303             "extra": "allow",
304             "validate_assignment": True,
305         }
306
307         # Temporary migration shim for legacy dict-style access (config.get("section",
{})).get(...)).
308         # Will be removed once all call sites (validators, segmenters, etc.) move to typed
309         # attribute access / the typed config. A nested *section* is returned as a plain
dict
310         # via model_dump, so legacy chained access keeps working ? returning the Pydantic
311         # section model instead breaks it (a model has no .get), which bit every validator
on
312         # real runs. Scalar leaves are returned as-is.
313         def get(self, key: str, default: Any = None) -> Any:
314             _MISSING = object()
315             val: Any = _MISSING
316             if key in type(self).model_fields or hasattr(self, key):
317                 val = getattr(self, key, _MISSING)
318             if val is _MISSING:
319                 # Legacy convenience: find a leaf key inside any nested section. Sections
are
320                 # discovered dynamically from the model fields, so a future top-level
section
321                 # (e.g. `audio`, `jobs`) is covered automatically without editing a
hardcoded list.
322                 for fname in type(self).model_fields:
323                     section = getattr(self, fname, None)
324                     if isinstance(section, BaseModel) and hasattr(section, key):
325                         val = getattr(section, key)
326                         break
327             if val is _MISSING:
328                 return default
329             if isinstance(val, BaseModel):
330                 return val.model_dump(mode="json")
331             return val

```

```

332
333     def __getitem__(self, key: str) -> Any:
334         val = self.get(key)
335         if val is None and key not in type(self).model_fields:
336             # Re-raise like a real dict for unknown top-level
337             raise KeyError(key)
338         return val
339
340
341     # Convenience type alias for callers that want to annotate
342     Config = AppConfig
343

```

4. user (2026-06-19T15:46:38.513Z)

```

1     # CLAUDE.md ? project memory / agent handoff
2
3     Local, AI-powered badminton (? racket-sports) highlight indexer + rally detector**:
4     FastAPI + SQLite + ffmpeg + GPU CV/AI, with paid Gemini as the cloud AI. Sibling
5     repos: `sports-data-collector`, a VLC-based `rally-annotator`.
6
7     ## Read these first
8     - docs/README.md ? doc index. docs/CODE_MAP.md ? code navigation + data
9       contracts (the cold-start map; read before touching code).
10    - Authoritative owned-model roadmap: docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md (+
the
11    live docs/NEXT_STEPS.md). These SUPERSEDE the 2026-06-07 strategy docs below where
they
12    disagree (licensing, Gemma-4 stance, base model, conversational-QA).
13    - Strategy (post-scaffolding): docs/COMMERCIALIZATION.md,
docs/PMF_MARKET_RESEARCH.md,
14    docs/PLATFORM_ARCHITECTURE.md, docs/DEFERRED_HARDENING_PLAN.md. Also
docs/ROADMAP.md
15    (offline-segmenter narrative) and docs/BACKLOG.md.
16    - Rally-detection quality track (design merged 2026-06-13, owner-gated, NOT
started):
17    docs/HOW_RALLY_DETECTION_WORKS.md (2-layer mental model) ?
docs/MULTI_SIGNAL_FUSION_PLAN.md
18    (fuse shuttle + person + audio cues; the held-shuttle bug is the already-built
rally_gate.py
19    "Gate A" not wired into production ? cheapest win) ? docs/EXPERIMENT_HARNESS.md
(A/B + shadow
20    eval so cues land flag-gated/default-OFF with zero regression) ?
docs/ROORA_LABELING_GUIDELINES.md
21    (v8 vendor labeling spec). docs/NEXT_STEPS.md has the prioritized pick-up for this
track.
22    - History: docs/archives/ ? ADRs (decisions/DECISIONS.md), research, and
23    checkpoints/ (latest: 2026-06-strategy-foundation/). Archive policy: only move
24    terminal-state (DONE/REJECTED) work; live docs stay at docs/ top level.
25
26    ## Current state (June 2026)
27    - Scaffolding complete; strategy foundation + owned-model roadmap merged. No
28    platform/owned-model implementation has started. A 2026-06-10 audit-driven hardening
29    sweep landed (licensing gate, detector keying, re-ingest safety, config, eval gate,
API
30    validation, reliability, CI + test coverage) ? see git history.
31    - Two tracks, both owner-gated: the committed next PLATFORM slice = async job
model
32    (DEFERRED_HARDENING_PLAN.md #16, single-tenant); the owned-model track starts at
a
33    greenlit Phase-0 milestone (OWNED_MODEL_IMPLEMENTATION_PLAN.md). Do not start
34    implementation without explicit owner approval.
35
36    ## Architecture in one breath
37    - Pluggable registries (ABC + Registry singleton): segmenters / providers /

```



```

validators
38      / sports / annotations ? see `backend/pipeline/segmenters/base.py`. *Future:*
39      `StorageBackend` + `ComputeBackend` registries (PLATFORM $3).
40      - `video_id` = MD5 of the file ? `backend/utils/hashing.py::compute_video_id` (one
helper).
41      - Typed config = `backend/config/` (Pydantic). DB schema evolves via the `PRAGMA
42      user_version` migration ladder in `backend/infrastructure/database.py`.
43
44      ## Committed guardrails (non-negotiable)
45      - **Paid LLMs (Gemini/OpenAI/Anthropic) = inference / serving / fallback ONLY ? never a
46      training data source** (terms/copyright). Commercial-clean licensing for EVERY
dependency ?
47      code, data, AND weights: **MIT / Apache-2.0 / BSD only** (ratified 2026-06-09, owner-
agreed).
48      - **Owned-model base:** **Qwen3-VL (Apache-2.0)** primary, InternVL3 (MIT) fallback;
**Gemma-4
49      is AMBER / bench-only** (Apache grant likely but Prohibited-Use posture unconfirmed ?
counsel
50      needed; do NOT build on it without sign-off). **Reject Gemma 3** (viral license).
**Exclude
51      minors** from the training pool; per-user training data needs a separate explicit opt-
in.
52      - **v1 = rally + highlight + history + correction loop** ; **gait/biometrics strictly
future**
53      (GDPR Art. 9).
54      - **Generic data schema** ; `coach ? athlete` is an **optional overlay** , not baked in.
55
56      ## Workflow conventions
57      - **Branch off `master` ; ONE PR per work item; NO PR stacking.** Don't open a PR unless
asked.
58      - Tests: `python run_all_tests.py` (offline, fully mocked). Python 3.8+ ; `ffmpeg`
required.
59      - Never push to the default branch directly; descriptive commit messages.
60      - **Substantial design/project work gets a tracked project doc** following
`docs/PROJECT_DOC_TEMPLATE.md` (status header · decisions-locked · progress tracker ·
definition-of-done; archived on completion). Precedents: `docs/RALLY_QUALITY_RESEARCH.md` $7,
`docs/GOLDEN_REGRESSION_FIXTURES.md`.
61      - ?? This dev container lacks GPU / torch / cv2 / numpy ? the **full pipeline can't run
here** .
62      Verify pure-logic paths and hand machine-side checks (real video, GPU/WSL suite) to
the owner.
63

```

5. user (2026-06-19T15:46:38.957Z)

```

1      # Next steps (live) ? owned-model + multisport
2
3      **Updated:** 2026-06-13 (**rally-detection quality track** : fusion + experiment-harness
design docs merged,
4      owner-gated ? see the dedicated section below, the desktop-agent pick-up). Prior:
2026-06-10 (audit?M0?Gemini-battery session; later same day the **UI-alpha track opened** :
5      ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-
first delivery ? see
6      `docs/UI_ALPHA_EXECUTION_PLAN.md` ; **PR-1 async jobs (#16) MERGED 2026-06-11, PR #87** ;
**PR-2
7      upload/download MERGED 2026-06-11, PR #99** ? live-E2E-verified (3-part chunked upload
MD5 byte-identity +
8      compile?browser-download); next: PR-3 identity. Same day: the **8?9 quality sweep
#90?#98 landed** ? ruff/mypy/
9      coverage-floor CI lanes, SQLite conn-close fix, config unknown-key warning, hybrid-twin
collapse into
10     `trajectory_hybrid.py`, WSL tilde+abspath live-path fixes; suite 388 green; LOVO 0.626
re-verified exact).
11     **Authoritative roadmap:**
12     `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+ ADR-008 for topology). **Latest blow-by-

```

```

blow:** memory
13     `current-state.md` / `session-handoff.md`. This file = the prioritized "what to do
next," refreshed each session.
14
15     ## ? Next-session / GPU-box pickup (2026-06-15)
16     **M0 in-container quick wins are COMPLETE:** ship-gate provisional blocker (**#170**),
train/eval split guard (**#171**), the **doc-consistency sweep** (**#174** ? corrected 16
drifted docs vs the code), and the **P3 person-feature fusion litmus machinery is already
present + suite-green** (`backend/eval/fusion_compare.py` + `ablation.py`: person-driven ?F1 /
bootstrap CI / paired sign test; the real-golden `skipif` litmus waits only on GPU-extracted
data). ? CI is red repo-wide due to a **GitHub Actions minutes/billing outage** (account-level,
not code) ? work is local-gated + `--admin`-merged until Actions is restored (Settings ? Billing
? Actions minutes / spending limit).
17
18     **Immediate next step ? the real P3 LOVO measurement, split GPU?CPU** (the dev box is
CPU-only: `torch-cpu`/`cv2` present but no GPU, and the corpus/videos/CSVs/labels are not here):
19     1. **[owner / GPU box]** `python -m backend.eval.fusion_golden --manifest
output/human_lovo_manifest.json --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-
data\features\2026-06-15-n6" ? produces `_golden_features.json` for the 6 golden videos. (GPU
+ WSL + videos + trajectory CSVs + `rallies.csv` labels are owner-side only.)
20     2. Artifacts land in the **shared cloud folder** `C:\Users\avidu\OneDrive\rally-
annotator\golden-data\` (OneDrive auto-syncs both ways). Convention + workflow:
**[`GOLDEN_DATA_SHARING.md`](GOLDEN_DATA_SHARING.md)**; tracked in **#175**.
21     3. **[CPU dev/agent session]** `python -m backend.eval.fusion_compare --features-dir
<shared>\features\2026-06-15-n6 --record` + `python -m backend.eval.ablation --features-dir
<shared>\... --granularity group` ? real person-feature ?F1; copy the JSONs into `output/` to
also pass the `skipif` litmus tests (`tests/test_ablation.py::test_litmus_reproduces_gate_a`).
22
23     **Also queued (owner-gated / pending input):**
24     - Ship-gate target calibration ? **#172** (owner decision; needs corpus growth).
25     - Tests reorganization ? blocked on the owner's grouping metric (provide it ? an agent
will do it).
26     - Restore CI ? the Actions outage is account-level (billing/minutes), not a repo/config
bug.
27
28     **Leaving (bulky / M2 ? no start without owner go):** ActionFormer (M0.4), event-index
DB migration (M0.1), Gemini-silver purge (M0.3), cost-to-Gen-2 benchmark + ByteTrack
deprecation, multisport, repo rename, PMF execution.
29
30     ## Where we are (one paragraph)
31     ADR-008 ratified ("repo split driven by productionization, not isolation"): training
lives in-repo (`training/`)
32     behind a CI-enforced import wall; the featurizer is single-sourced
(`backend/features/rally_seq.py`, train==serve);
33     the **Gen-0 harness ships** (`python -m training.gen0.harness --manifest
output/human_lovo_manifest.json --floor
eval_baselines/heuristic_lovo_n6.json --version <semver>`) and produced artifact v0.1.0
(LOVO **0.626**, floor passed,
34     sign test 5/6 wins p=0.109 ? honestly not significant, ship=False). The **n=6 owned
corpus is in the collector**
35     (262 rallies, Proprietary, commercial export = owned data only after the ShuttleSet
reclassification). The **Gemini
36     battery is done** (see table) ? the moat is quantified.
37
38
39     ## The quality table that now drives strategy (IoU 0.5 vs human golden labels)
40     | Path | Mahadevpura (clean-ish) | Kushagra (multi-court) |
41     |---|---|---|
42     | Heuristic (velocity windowing) | 0.657 | 0.157 |
43     | **Gen-0 owned head** (LOVO, n=6) | 0.744 | 0.561 |
44     | Two-stage WASB+Gemini refine | 0.83 | ? |
45     | **Gemini wrapper** (`gemini-flash-latest`) | **0.93** | **0.66** |
46
47     **Reading:** clean footage is commoditized (serve it with Gemini for cents). **Multi-
court drops the commodity
48     wrapper to 0.66 ? that is the owned model's ship target** (its FPs are background-game

```

pickups + dead-time merges,
49 the exact contrast-metric confound). Gen-0 is 0.10 behind with only n=6 ? a corpus-
growth-sized gap, not an
50 architecture-sized one. Two-stage refine = the cost-efficient serving path for LONG
tapes (uploads candidate clips
51 only); wrapper 503-flakiness (observed all session) makes the provider-fallback registry
load-bearing.
52 Baselines tracked in `eval\_baselines/` (heuristic_lovo_n6.json,
gemini_wrapper_2026-06-10.json).
53
54 ## Rally-detection quality track (design merged 2026-06-13, owner-gated; implementation
in progress)
55 **This is the track a desktop/local agent is slated to pick up.** Design docs landed (PR
#112 + session). All implementation is **owner-gated** (per explicit in the plan); each step =
ONE PR off `master`.
56
57 **Hardening loop (T1-T5 positive tests + audit + finale) COMPLETE (2026-06-14, PRs
#156/#157/#160/#161/#163 + #165 + final records).** See archived
`docs/archives/CODE\_AUDIT\_AND\_TEST\_HARDENING-COMPLETED-2026-06-14.md` (Living Log with every
pre/post/review) and `docs/archives/HARDENING\_LOOP\_HANDOFF-COMPLETED-2026-06-14.md` (full rules
+ STATE). Top-level now pointer stubs per #165 review. All 5 items + verification + archive
done; clean slate.
58
59 - **Docs:** `docs/MULTI\_SIGNAL\_FUSION\_PLAN.md` (shuttle + person + audio, one fusion
layer) .
60 `docs/EXPERIMENT\_HARNESS.md` (A/B + shadow + promotion gate, zero-regression) .
61 `docs/ROORA\_LABELING\_GUIDELINES.md` (v8) . `docs/HOW\_RALLY\_DETECTION\_WORKS.md`
(2-layer mental model).
62 - **Key finding (no re-investigation needed):** the owner's "held-shuttle counted as a
rally" bug is **NOT a missing capability** ? `backend/pipeline/detectors/rally\_gate.py` already
implements the net-crossing **"Gate A" (`apply\_gate(..., min\_crossings=2)`) that kills
service-feed/held-shuttle windows, but it is **only called from eval drivers** (and there was no
`min\_crossings` knob in `IndexingConfig`). **Wiring Gate A into the live segmenter (now via
fusion_hybrid) was the single cheapest, highest-confidence quality win.** (P2 FusionSegmenter
specifics: registry `fusion\_hybrid`, **default-OFF**, seeds from substrate, persists
`net\_crossings` + optional person features, optional served Gate A/B via config knobs;
adversarially reviewed; suite green. See `tests/test\_served\_gate\_a\_regression.py` for the honest
finding that on production windowing Gate A is neutral/negative in some cases, hence kept opt-in
with knob for safety, and the leaderboard revert note.)
63 - **Current status (2026-06-14):** P1 (person cue extraction to reusable
`PersonDetector`) + P2 (FusionSegmenter + Gate A signal + served handoff gating) + experiment
harness P1 DONE (adversarial review + NO-REGRESSION; suite green). **P3 next (this PR + follow-
ups):** learned fusion (person features ? harness `compare` LOVO lift on the 6 golden videos;
eager-person-compute optimisation); then P4 audio via hit_detector.
64 - **Pick-up order (each = ONE PR off `master`, owner approval first, flag-gated default-
OFF, promote only on measured LOVO):**
65 1. (P2 first-step, largely done via fusion_hybrid) Wire Gate A into production served
path + config knob (min_crossings); apply in hybrids at runtime. (Now available opt-in; served
neutral on production windowing per litmus ? kept opt-in with knob for safety.)
66 2. P3 first step (this work item): harness `compare` LOVO litmus for person-on variant
(players_in_court + two_sided + colocation etc. from `fusion\_features`) on the 6 golden; add the
eager compute optimisation note/landing. Pure + harness (testable here).
67 3. P3/P4 continuation: audio cue integration + full learned promotion if lift proven.
68 - **Discipline (from EXPERIMENT_HARNESS.md):** every new cue **flag-gated, default
OFF** ; promote to default **only on measured LOVO lift** through `run\_regression.py` (exit
0/1/3); pinned `CONTROL` ? rollback = config flip. Harness re-uses ~80% existing code.
69 - ? Dev container no GPU/torch/cv2 ? Gate A, person cue logic, harness, and pure fusion
paths are testable here; real-video / golden LOVO / ablation confirmation is owner-side (GPU/WSL
machine).
70 - **External due-diligence review inputs (2026-06-17) ? PICK UP AFTER current ideas are
exhausted; full details + citations in [`RALLY\_QUALITY\_RESEARCH.md`](RALLY_QUALITY_RESEARCH.md)
\$5.6.** An owner-commissioned 31-reference review of the quality report *validated* the
methodology and surfaced these **future** levers (research inputs, not active work, not new
claims; verify citations before any submission):
71 1. **"First-mile" paper framing (highest-value):** stroke-classifiers (BST/DSTA) + the

big datasets (ShuttleSet ~36k strokes, Fine-Badminton ~27k actions) all **assume a pre-trimmed rally clip already exists** ? we solve the ignored prerequisite. Pitch the paper as "the missing untrimmed-video preprocessing pipeline that lets BST/MonoTrack run on raw footage."

72 2. **Ground R2 in Action-Spotting / Precise-Event-Spotting**
(SoccerNet/TennisSet/T-DEED) ? turns R2 into a defensible publishable contribution.

73 3. **Add domain baselines for any submission:** MonoTrack + BST on ShuttleSet / Fine-Badminton (we have only heuristic + Gemini today).

74 4. **Prefer TemporalMaxer over ActionFormer** for the M0.4 scorer swap (parameter-free max-pool, ~2.8x fewer GMACs, small-N robust) ? see "M0.4 next increment" below.

75 5. **Court polygons ? pose ? the rally-END lever:** masking on-court players can resolve *occluded end-of-rally states* (the #1 remaining gap vs Gemini), not just spectator noise.

76 6. **Name the eval?serve lesson "pipeline distribution skew"*** ? a citable paper "lesson."

77

78 **Discipline reminder:** owner approval before starting any owner-gated item; one PR per step; babysit the PR (poll/fix/reply) until merge observed, *then* advance.

79

80 ## Prioritized next steps

81 1. **[owner, in progress] GROW THE GOLDEN CORPUS** ? the single highest-leverage move; every lever below feeds on it.

82 Priority order for new videos: *(a) multi-court badminton* (the target distribution AND where the ship target

83 lives), *(b) ?3 matches per new sport ? table tennis first* (WASB transfers at F1 0.909; pickleball labels are

84 corpus-gold but not trainable until a clean MIT/Apache ball detector exists), (c) capture variety per

85 `VIDEO\_RECORDING\_GUIDELINES.md`; **adult sessions only for the training pool** (consent guardrail). Per video:

86 label ? `curate import-local --normalize --license Proprietary --fps <true fps>` ? re-run the Gen-0 harness.

87 The paired sign test needs n: at n=10 with 9 wins, p?0.011 ? significance is reachable this month.

88 2. **[owner decision] Set the ship-gate target** ? now informed: floor = heuristic 0.480 (necessary), **multi-court

89 reference = wrapper 0.66** (the number that makes the owned model worth serving), clean-footage ceiling = 0.93

90 (not the target ? Gemini serves that tier). Suggested shape: "LOVO F1@tIoU0.5 ? 0.70 on multi-court folds with

91 paired-sign p<0.05 vs heuristic" ? owner to ratify/adjust.

92 3. **M0.4 next increment (\$0/local):** swap the Gen-0 LogReg for a lightweight temporal head (MIT, minutes on the

93 2070S ? NOT the heavyweight Gen-2 stack) inside the existing harness; per-video threshold calibration (the

94 LOVO-vs-oracle gap is visible per fold in the harness output). **Prefer TemporalMaxer** (parameter-free

95 max-pool, ~2.8x fewer GMACs, small-N robust) over ActionFormer/ASFormer per the 2026-06-17 external review

96 (`RALLY\_QUALITY\_RESEARCH.md` \$5.6) ? better fit for the cheap/local/fast mandate.

97 4. **Multisport thread:** ingest Roora pickleball/TT CSVs (`import-local --normalize`); merged-prototype LOVO across

98 badminton+TT once TT has ?3 matches; identify a clean pickleball ball detector. ? TT-on-WASB serving stays gated

99 by C3 (provenance multiplies per sport).

100 5. **PMF validation (cheaper than any engineering month):** C13 academy interviews (+ the two added questions:

101 "multi-court footage nobody watches?" / "pay per-match to index it?"); camera pilot at the warmest 2-3 academies

102 (consent at capture; demo served via the Gemini path or human-polish ? the reel pattern exists:

103 `output/MahadevpuraSingles\_highlights.mp4`). **Field kit COMPLETE (2026-06-12, owner ratified the

104 physical-visit approach; reframed product-feedback-first per owner spec):**

105 `docs/PRODUCT\_FIELD\_ASSOCIATE\_FLYER.md` (the recruiting flyer ? owner fills ?/contact placeholders and recruits) +

```

106     `docs/FIELD_VISIT_PLAYBOOK.md` (the hired associate's visit playbook: coach + student
script, demo-after-Q6
107     rule, ground rules, Phase-2 record-and-show-back stretch goal) +
`docs/FIELD_VISIT_ANSWER_SHEET.md` (per-visit
108     capture form incl. demo-reaction + student blocks) + `docs/PMF_FIELD_SIGNALS.md`
(G/A/R anchors + **pre-registered
109     decision rules** ? PROCEED / PIVOT A-C / FALSIFIED thresholds fixed before the first
visit).
110 6. **Housekeeping:** ~rotate the Gemini API key~ **DONE 2026-06-10** (owner-verified
working; confirm old-key
111     deletion in AI Studio); **rename repo off "badminton"** (owner action,
112     in BACKLOG); collector=SoR confirmation; off-peak niceties: none pending ? the
battery completed.
113
114     ## Pointers / artifacts
115     - Roadmap: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` . Topology: ADR-008
(`docs/archives/decisions/DECISIONS.md`) .
116     Study: `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md` . Quality history:
117     `docs/archives/quality-iterations/2026-06-multivideo-lovo/`.
118     - Harness: `training/gen0/harness.py` (one command ? train ? LOVO ? gate ? artifact
bundle; weights gitignored,
119     manifest reviewable). Featurizer: `backend/features/rally_seq.py` (FEATURE_SCHEMA v1).
120     - Corpus: collector DB (system-of-record, 6 matches/262 rallies) . eval manifest
`output/human_lovo_manifest.json`
121     (gitignored) . durable trajectories + Roora files: `C:\Users\avidu\Projects\Annotation
Setup\Collect\Trajectories\` .
122     labels: `...\Collect\Golden Labelled\`.
123     - Gemini ops learnings: serial uploads (`indexing.ai_max_workers=1`) + chunk re-encode
124     (`indexing.ai_chunk_scale_width=1280`) + 6-attempt exponential generate retry ? a
cold/prepaid project burst-throttles
125     with a MISLEADING "credits depleted" 429; ramp gradually.
126     - ? No owned-model implementation beyond greenlit M0.x until owner greenlights the next
milestone. Committed next
127     PLATFORM slice = async job model (single-tenant).
128

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6. user (2026-06-19T15:46:39.399Z)

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1     - [ROORA_LABELING_GUIDELINES.md](ROORA_LABELING_GUIDELINES.md) ? Latest (v8) vendor
labeling guidelines mirrored from the collector's `docs/annotation/`, with the fusion-driven
court-calibration / singles-doubles additions marked `? ADDED (fusion)`.
2     - [RALLY_QUALITY_RESEARCH.md](RALLY_QUALITY_RESEARCH.md) ? **Foundation doc** for
closing the local-vs-Gemini rally-detection gap: the measured gap + the code-grounded over-merge
diagnosis ("shuttle moving ? rally"), an inventory of the eval/experiment harness + its gaps,
the strengthened evaluation framework (segmental Edit/F1@k that make over-segmentation first-
class, golden-corpus + active learning, Gemini-agreement as a *shadow* signal), an
**adversarially-verified, cited external-research synthesis** (TAL/TAS boundary heads, racket-
sports rally-state cues, small-data, eval rigor), a **prioritized cheap?durable roadmap with
falsifiable success criteria**, and ($7) a **tracked project plan** ? work-item checklist,
sequencing, the single quality **number** attributed to the effort, and a **definition of done**
(project completes when all tiers land + the number is recorded). The plan that the
[QUALITY_ITERATIONS.md](QUALITY_ITERATIONS.md) log executes against; ties together
EXPERIMENT_HARNESS / MULTI_SIGNAL_FUSION_PLAN / ANALYZERS_AND_RECONCILERS /
OWNED_MODEL_IMPLEMENTATION_PLAN.
3     - [RALLY_DETECTION_GAP_CLOSURE.md](RALLY_DETECTION_GAP_CLOSURE.md) ? **Tracked execution
doc (2026-06-18)** for closing the remaining rally-detection accuracy gap **per video, before
launch**, driven by the growing golden corpus. First ground-truth clip (GX010141) reframed it:
boundaries are tight (R2/R3/R4 holds) ? the gap is **detection**: (G1) recall (local & Gemini
both miss ~half on hard clips, and *different* rallies ? complementary), (G2) precision (local
over-fires), (G3) **rally-END over-extension on the quick-return-after-point** (shuttle still
moving ? needs a *player-state* signal). Levers (owner-gated): player-kinematics + shuttle-in-
hand / sustained-invisibility end-rule with reverse-timeline backtrack, `rally_gate` Gate A
wiring, complementary fusion. **Peer** of `RALLY_QUALITY_RESEARCH.md` (execution tracker, not
research); $7 is the source of truth.
4     - [RALLY_DETECTION_QUALITY_REPORT.md](RALLY_DETECTION_QUALITY_REPORT.md) ? **Technical-

```

paper checkpoint (2026-06-17)** of rally-detection quality: the honest measured state (over-segmentation milestone tolerance-F1 0.09170.323 at n=11, p=0.002; merge-rate 0.69470.150; first real-footage ground-truth at-par with START at Gemini parity), the task-faithful R2 metric, a detailed Gemini comparison, next steps + expected wins, areas yet to be explored, and a publishability / path-to-submission verdict. Renders to a polished PDF; the externally-shareable "toward a paper" summary of `RALLY\_QUALITY\_RESEARCH.md` + `QUALITY\_ITERATIONS.md`.

5 - EVALUATION.md ? How we measure and improve quality.

6 - EXPERIMENT_HARNESS.md ? Design (P1 SHIPPED) for A/B + shadow + SxS experimentation on rally-detection variants with **zero production regression** ? experiments are decoupled from deploy (new cues default OFF, promote only on measured LOVO lift, rollback = flip a flag). Composes the existing `backend/eval/` machinery (`calibration`, `regression`, `metrics`); implemented in `backend/eval/experiment.py`.

7 - QUALITY_ITERATIONS.md ? **Living log** of rally-detection quality iterations (hypothesis ? what shipped ? measured/expected effect ? lesson). The reverse-chronological record; terminal-state snapshots get archived under `archives/quality-iterations/`.

8 - ANALYZERS_AND_RECONCILERS.md ? Design (owner-gated, NOT started): a **signal-fusion framework** for rally detection ? producers emit confidence-scored **signals** at three temporal **scopes** (frame detectors / window analyzers / session analyzers), a learned **scorer** weights them, and an auditable **report-first reconciliation** pass lints the decision against the evidence. The spine: **no special-casing in the decision path** (signals?scorer, never `if/else`). Worked examples: a service-fault analyzer, a warm-up/practice span detector, and consuming a per-frame shuttle-state detector. A peer to `EXPERIMENT\_HARNESS.md` and `MULTI\_SIGNAL\_FUSION\_PLAN.md` (ADR-007 modular rally layer).

9 - PERSONALIZATION_PLAN.md ? Design (owner-adopted **hybrid**; Phase-0 landing) for the **per-user personalization** feature on a generalization-first baseline: a thin per-user **tuning** layer gated by the honest small-N statistics (a **min-N promotion floor** + a **structural-guard exemption**, never a service gate) + a **contribution flywheel** (free-tier videos pool into the owner-trained baseline; **"a tool that gets better the more you help it"**). Records the **locked owner decisions** (identity, storage, `min\_n` semantics, Gate-A default, free-tier pooling). Built so far: per-user promotion gate (#141) + held-out-USER learning curve (#142).

10

11 - GOLDEN_REGRESSION_FIXTURES.md ? **Design** (owner-gated, NOT started): **video-free, non-attributable regression fixtures** so a clean clone runs the rally-seg suite with **no video/GPU/DB**. Two layers ? the post-detector freeze + **dual gates** (characterization + quality-floor) reusing the `backend/eval/` stack, and the **SEALED-FIXTURES** privacy filter (de-attribution-at-source, tiered public/key-gated, shuttle-only public tier). Carries a cited **IN/US/EU legal memo** (derived-data IP/privacy), a threat model, and a **\$7 progress tracker + definition-of-done**. Extends GOLDEN_DATA_SHARING.md.

12

13 ## Product / platform plans

14 - DESKTOP_APP_PLAN.md ? **Tracked scoping doc** (2026-06-18, owner-gated, `[design]` ? NOTHING built): **package the local pipeline as a cross-platform desktop app** for non-technical players ? plug in the camera's USB/SD card, click once, get back **only the highlight reels**, fully on-device (no upload, no original-video bloat). The centerpiece is the **de-WSL native-compute port** (run WASB natively per-OS: Linux/Windows CUDA · macOS MPS/CPU ? drop WSL2), and its **make-or-break unknown**: is CPU acceptable on a typical **no-GPU** enthusiast laptop (WASB is ~3.4x real-time on a laptop 2070S)? \$7 tracker = **P0 compute-feasibility spike** ? P1 native detector ? P2 app shell ? P3 packaging/installers. Realizes the **L0 fully-local** tier of VIDEO_LOCALITY_MODEL.md; is the **LocalDesktop` ComputeBackend** of PLATFORM_ARCHITECTURE.md §3.2; the smaller/export-clean owned model (OWNED_MODEL_IMPLEMENTATION_PLAN.md) is one answer to the compute crux.

15

16 ## Practical / ongoing

17 - TRACKNET_WSL_SETUP.md

18 - Golden set docs (`GOLDEN\_SET`)

19 - Evaluation harness and calibration drivers (in `backend/eval/`)

20 - GOLDEN_DATA_SHARING.md ? How the GPU box and CPU dev/agent sessions share golden artifacts (trajectory CSVs, `\_golden\_features.json`) via a OneDrive folder; the GPU-extract ? CPU-litmus split. See **##175**.

21

```

22     ## Conventions & templates
23     - [PROJECT_DOC_TEMPLATE.md](PROJECT_DOC_TEMPLATE.md) ? Reusable skeleton for a **tracked
project doc**: status header · decisions-locked · progress tracker (??/?) · definition-of-done
· archived on completion. Standardizes the emergent pattern in `RALLY_QUALITY_RESEARCH.md` §7
and `GOLDEN_REGRESSION_FIXTURES.md`. Use for any substantial design/project work.
24
25     ## Historical documents
26     Completed plans, old research, ADRs, and checkpoints live in [`archives/`](archives/).
27
28     - [CODE_AUDIT_AND_TEST_HARDENING.md](CODE_AUDIT_AND_TEST_HARDENING.md) ? **COMPLETED**
(2026-06-14; T1?T5 + full loop + verification per #162). Living record of the June 2026 plumbing
/ contract / architectural audit, 626-test infrastructure, positive test plans for the 5
identified issues, multi-layer regression protocol (including ablation/experiment layer), and
pre/post-change verification checklist. See `docs/archives/CODE_AUDIT_AND_TEST_HARDENING-
COMPLETED-2026-06-14.md` and `docs/archives/checkpoints/2026-06-14-hardening-loop-complete.md`
for the final archived state. (Top-level is now pointer stub per review; full content in
archives/.)
29
30     See `archives/README.md` for the umbrella policy and `archives/past_projects/README.md`
for classification of small tech debt / starter projects.

```

7. assistant (2026-06-19T15:46:42.845Z)

Now let me verify the seeds. SEED 1 says the dev container CAN run the full pipeline. Let me verify cv2/numpy/torch import and check the actual repo state, and look at the NEXT_STEPS.md statements about CPU-only.

8. user (2026-06-19T15:46:51.524Z)

```

3.13.13 (tags/v3.13.13:01104ce, Apr  7 2026, 19:25:48) [MSC v.1944 64 bit (AMD64)]
---numpy---
numpy 2.4.6
---cv2---
cv2 4.13.0
---torch---
torch 2.6.0+cu124 cuda True

```

9. assistant (2026-06-19T15:46:55.366Z)

SEED 1 confirmed: numpy, cv2, and torch all import fine, and `torch.cuda.is\_available()` returns True (GPU present). This directly contradicts CLAUDE.md line 61-62 and several NEXT_STEPS.md claims. Let me verify the seed about local windowing being default-ON is reflected correctly, and check whether NEXT_STEPS / README make stale claims about it. Let me look at git history and the relevant docs more carefully.

10. user (2026-06-19T15:46:57.025Z)

```

ebe2e6d fix(gemini): surface failed chunks loudly (no more silent partial reels)
a3b4954 feat(runner): --min-rally-duration flag (surface the long-rally reel filter on the CLI)
(#238)
a240221 fix(test): ablation litmus uses canonical experiment._is_fusion_schema (unbreaks the
suite) (#237)
d749fb9 test(eval): re-ground served Gate-A litmus on the present (multicourt) golden set (#236)
ea6b2b8 docs(desktop): scope a cross-platform desktop app + the de-WSL native-compute port
(#219)
20f8aa3 feat(eval): P7 long-rally (>5s) quality lens + reel knob (both default-OFF) (#220)
5f555cf refactor(eval): extract one shared golden-feature loader behind the three load_videos
(#235)
5409fc4 refactor(api): extract job-worker drain/wake loop from main into backend.api.job_worker
(#234)
5c46ebc refactor(eval): split real-fixture subsystem into golden_real_fixtures (pure move)
(#233)
97fa187 refactor(db): route bool-returning single-statement writers through _execute_write
(#232)

```

```
ae10447 refactor(eval): extract _build_preset(args, side) from rally_seg_eval.main() (#231)
1e324ea refactor(motion): extract detection + DB-populate halves from process_video (#230)
53b9cef docs(code-map): point UPLOAD endpoints at backend/api/uploads.py (post #228) (#229)
e67add1 refactor(api): extract chunked-upload island from main into backend.api.uploads router
(#228)
8092a39 refactor(segmenter): decompose yolo_hybrid process_video into phase helpers (#227)
```

11. user (2026-06-19T15:46:58.891Z)

```
15:## ? Next-session / GPU-box pickup (2026-06-15)
18:**Immediate next step ? the real P3 LOVO measurement, split GPU?CPU** (the dev box is CPU-
only: `torch+cpu`/`cv2` present but no GPU, and the corpus/videos/CSVs/labels are not here):
19:1. **[owner / GPU box]** `python -m backend.eval.fusion_golden --manifest
output/human_lovo_manifest.json --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-
data\features\2026-06-15-n6" ? produces `*_golden_features.json` for the 6 golden videos. (GPU
+ WSL + videos + trajectory CSVs + `.rallies.csv` labels are owner-side only.)
21:3. **[CPU dev/agent session]** `python -m backend.eval.fusion_compare --features-dir
<shared>\features\2026-06-15-n6 --record` + `python -m backend.eval.ablation --features-dir
<shared>\... --granularity group` ? real person-feature ?F1; copy the JSONs into `output/` to
also pass the `skipif` litmus tests (`tests/test_ablation.py::test_litmus_reproduces_gate_a`).
69:- ? Dev container no GPU/torch/cv2 ? Gate A, person cue logic, harness, and pure fusion paths
are testable here; real-video / golden LOVO / ablation confirmation is owner-side (GPU/WSL
machine).
```

12. assistant (2026-06-19T15:47:05.546Z)

SEED 1 is strongly confirmed. NEXT_STEPS.md has multiple stale "no GPU/torch/cv2" / "CPU-only" / "torch+cpu" claims that directly contradict the environment (torch 2.6.0+cu124, cuda True, cv2 4.13.0, numpy 2.4.6).

Now let me verify the seed about local windowing default-ON. The seed says to flag any meta doc calling local windowing not-started / default-OFF / scaffolding-only. Let me check what NEXT_STEPS.md and README say about the quality track / Gate A / windowing status, and verify against code. Note the seed's specific subject is the cap=6 + R4 milestone (max_window_duration, window_inactivity_end, etc.), which I've confirmed default-ON in models.py.

Let me check the "implementation in progress" vs "NOT started" language and the served Gate A status more carefully.

13. user (2026-06-19T15:47:07.590Z)

```
4:owner-gated ? see the dedicated section below, the desktop-agent pick-up). Prior: 2026-06-10
(audit?M0?Gemini-battery session; later same day the **UI-alpha track opened**:
23:**Also queued (owner-gated / pending input):**
54:## Rally-detection quality track (design merged 2026-06-13, owner-gated; implementation in
progress)
55:**This is the track a desktop/local agent is slated to pick up.** Design docs landed (PR #112
+ session). All implementation is **owner-gated** (per explicit in the plan); each step = ONE PR
off `master`.
62:[Omitted long matching line]
64:- **Pick-up order (each = ONE PR off `master`, owner approval first, flag-gated default-OFF,
promote only on measured LOVO):**
68:- **Discipline (from EXPERIMENT_HARNESS.md):** every new cue **flag-gated, default OFF**;
promote to default **only on measured LOVO lift** through `run_regression.py` (exit 0/1/3);
pinned `CONTROL` ? rollback = config flip. Harness re-uses ~80% existing code.
78:**Discipline reminder:** owner approval before starting any owner-gated item; one PR per
step; babysit the PR (poll/fix/reply) until merge observed, *then* advance.
81:1. **[owner, in progress] GROW THE GOLDEN CORPUS** ? the single highest-leverage move; every
lever below feeds on it.
```

14. user (2026-06-19T15:47:07.958Z)

```
1      # Rally-detection quality iterations ? living log
2
```



```

3      **Purpose.** A running, reverse-chronological log of how we evolve rally-detection
quality: each
4      iteration's hypothesis, what shipped, the measured (or expected) effect, and the lesson.
This is the
5      **live** continuation of the historical trail in
6      [archives/quality-iterations/](archives/quality-iterations/README.md) ? per the
archive policy
7      (CLAUDE.md), **terminal-state** (DONE/REJECTED) work gets a dated subdir there; this
file stays at
8      docs/ top level and keeps growing, then its entries graduate to the archive when they
conclude.
9
10     **Methodology (non-negotiable).** Every quality change rides the experiment harness
11     (EXPERIMENT_HARNESS.md)(EXPERIMENT_HARNESS.md): new cues land **flag-gated, default
OFF**; they're
12     graded **offline against golden labels** (per-video + LOVO-honest); promotion to default
happens **only
13     on measured lift** through the run_regression.py gate; **rollback = flip a config
flag, never
14     git revert. **Cues are designed in
[MULTI_SIGNAL_FUSION_PLAN.md](MULTI_SIGNAL_FUSION_PLAN.md); scoring
15     in [EVALUATION.md](EVALUATION.md); the 2-layer model in
16     [HOW_RALLY_DETECTION_WORKS.md](HOW_RALLY_DETECTION_WORKS.md).
17
18     **Golden corpus = the 6 labelled matches** in output/human_lovo_manifest.json (owner-
confirmed
19     2026-06-13: use all 6 ? adarsh_avi_singles, kushagra_singles, largetest_doubles,
gbaaddy,
20     testlarge_short, mahadevpura_singles). The model trained on these is what we run
alongside the existing
21     detector on NEW footage for the side-by-side review.
22
23     ---
24
25     ## Baselines that drive strategy (IoU 0.5 vs human golden labels)
26
27     | Path | Mahadevpura (clean-ish) | Kushagra (multi-court) |
28     |---|---|---|
29     | Heuristic (velocity windowing) | 0.657 | 0.157 |
30     | Gen-0 owned head (LOVO, n=6) | 0.744 | 0.561 |
31     | Two-stage WASB+Gemini refine | 0.83 | ? |
32     | Gemini wrapper (gemini-flash-latest) | 0.93 | 0.66 |
33
34     **Reading:** clean footage is commoditized (serve via Gemini); **multi-court drops the
commodity wrapper
35     to 0.66 ? that is the owned model's ship target. **Baselines tracked in
eval_baselines/. The owner's
36     observed false positive ? a player **holding/flicking** the shuttle between points
counted as a rally ?
37     is the precision bug the fusion gates target.
38
39     ---
40
41     ## Current track (live) ? newest first
42
43     ### P7 ? Long-rally (>5s) quality lens (eval) + reel-selection knob ? MEASURED, default-
OFF *(2026-06-18)*
44     **Hypothesis (owner).** Local OVER-FIRES, and its false positives are mostly SHORT
(motion blips,
45     background-court fragments on multicourt). Filtering the eval to **long rallies (>5s)**
? the
46     eye-catching highlights that carry the product ? should strip most short FPs and reveal
local's real
47     long-rally quality, answering the North-star "do we beat Gemini on the single-match
long-rally happy

```

```

48     path?". **Duration** (end?start), NOT shot-count, is the filter: golden `shots_count` is
blank and the
49     no-AI path reports it "Unknown", whereas duration comes from the start/end labels we
already trust.
50
51     **What shipped (both default-OFF, zero behaviour change at the defaults).**
52     - **Eval lens** ? `rally_seg_eval.py --min-duration <s>` filters BOTH predictions AND
golden GTs to
53     rallies longer than `s` *before* matching, consistently across the tIoU and R2
tolerance paths (a
54     FILTER, not a new metric). Plus `--stratify` / `--long-tau` (default 5s): a `{all vs
long} x {single
55     vs multicourt}` aggregate table. Court type = the committed `MULTICOURT_CLIPS` set
(the manifest is
56     gitignored, so the set is the auditable source), overridable by a manifest
`court_type` field; kept
57     in sync with [`GOLDEN_VIDEOS.md`](GOLDEN_VIDEOS.md).
58     - **Product knob** ? `stitching.min_rally_duration` (s, 0 = OFF) drops short rally pairs
before the
59     reel compile in `tools/generate_highlights.py` ? a cleaner reel-selection filter than
60     `min_shot_count` (which leans on the "Unknown" shot_count).

```

15. assistant (2026-06-19T15:47:14.965Z)

Now let me look at the QUALITY_ITERATIONS R1 milestone entry to confirm the cap=6/R4 default-ON status is documented as shipped (this gives me the corroborating evidence that the windowing milestone is default-ON, not a design proposal). Let me also check whether the "Current state (June 2026)" / scaffolding language in CLAUDE.md is in my slice and whether it's stale. The seed is specifically about local windowing default-ON, so let me focus on whether any meta doc calls windowing not-started/default-OFF.

The CLAUDE.md "Current state (June 2026)" section says "Scaffolding complete... No platform/owned-model implementation has started" ? that's about the platform/owned-model tracks (owner-gated roadmap intent), which I must NOT relitigate. But I should check if CLAUDE.md anywhere calls the local windowing track not-started.

Let me look at the rally-detection quality track section in CLAUDE.md (lines 16-21) and check whether Gate A "not wired into production" is still accurate, and whether windowing is described.

16. user (2026-06-19T15:47:16.978Z)

```

19: "Gate A" **not wired into production** ? cheapest win) ? `docs/EXPERIMENT_HARNESS.md` (A/B
+ shadow
33: greenlit Phase-0 milestone (`OWNED_MODEL_IMPLEMENTATION_PLAN.md`). **Do not start

```

17. user (2026-06-19T15:47:18.665Z)

```

62:**Measured (n=15 golden, milestone cap6+R4, LOCAL vs golden; reproduce: `python -m
backend.eval.rally_seg_eval --stratify`):**
87: mahadevpura_1's 63s blob, GX010141 14.8s) and cap=6 truncates genuinely-long rallies so
the strict
104:### R1 ? MILESTONE LAUNCHED (cap=6 + R4 flipped default-ON) ? re-verified at n=13
*(2026-06-18)*
105:The R1 milestone is **shipped** (PR #204): the w1 cap=6 + R4 inactivity-end defaults are now
ON.
107:the **[R1 Milestone Launch Report](R1_MILESTONE_LAUNCH_REPORT.md)** (per the Launch-Report
convention).
109:**Re-verified on the grown n=13 corpus** (the 2 new Boxhill clips GX010137/GX010141) ? the
milestone
114:| baseline ? milestone tolF1 | ?+0.232, 10/0, p=0.002 | **?+0.222, 12/0/1, p=0.0005** |
120:milestone 0.474. Independently reproduced by a 2nd harness (#214 `lovo_resweep.py`, matches
to 3 dp;
121:LOVO picks cap=6 **13/13** on tIoU) and survived a 5-lens adversarial workflow (every L00

```

fold clears

129:The one clip the cap6+R4 milestone still fails is the ****continuously-active blob****
(mahadevpura_1:

146:****MEASURED** (n=11, on the cap6+R4 milestone, R2 `tolF1`):******

148:| clip | milestone | + R5 | windows (milestone ? +R5) |

160:it on selectively for continuously-active footage. The R1 net over-seg guard confirms it:
milestone?+R5 net

163:### R1 ? milestone evidence (cap=6 + R4) + the net over-seg promotion guard *(2026-06-17)*

164:The R1 milestone is the smallest safe local-windowing flip: ****W1 bounded windowing cap=6 + the R4**

166:the full ablation ladder baseline?milestone under the ****R2**** `tolF1` on ****n=11**** with honest paired

167:sign-tests, and (b) ships the ****net over-seg promotion guard**** that unblocks the milestone.

175:| + cap=6 (R3) | 0.276 [0.174, 0.380] | 0.407 | 5.7 s |

176:| ****+ R4** **inact_end=1.5** (milestone)****** | ****0.323** [0.210, 0.433]****** | ****0.440**** | ****3.8 s**** |

179:- ****baseline ? milestone:**** ?`tolF1` ****+0.232**** [+0.153, +0.304], ****sign 10/0/1, p=0.002**** (the 1 tie is the

181:- ****cap=12 ? cap=6**** (R3 was means-only ? now confirmed): ?`tolF1` ****+0.108**** [+0.070, +0.154], ****sign 10/0/1,**

182: p=0.002******. ****cap=6 beats cap=12, significantly.****

183:- ****cap=6 ? +R4:**** ?`tolF1` ****+0.047**** [+0.022, +0.072], ****sign 8/0/3, p=0.008**** (reproduces the R3/R4 entry).

189:and the raw merge ****count**** is non-monotonic. The milestone trips strict on 10/11 videos, ****yet it strictly**

205:****R1 verdict:**** the milestone (cap=6 + R4) clears the refined (net-guard) promotion bar ? a significant `tolF1`

207:(`indexing.max\_window\_duration: 6`, `window\_inactivity\_end: 1.5`, `inactivity\_rally\_thresh: 0.20`,

216:- ****R3 ? re-scope the cap (12 ? 6 s).**** The 12 s cap was too coarse. Sweep (W1, no R4):
`tolF1` cap=6 ****0.273****

217: > cap=8 0.234 > cap=12 0.171. So R1's recommended cap is ****6 s****, not 12. (Means-based lead across the

219:- ****R4 ? rally-END inactivity trim**** (`window\_inactivity\_end`, ****default-OFF****). The measured gap is the

227: ****MEASURED** (n=11, on top of W1 cap=6; sweet spot ?_end-window=1.5 s, rally_thresh=0.20, min_density=0.30):******

229: | metric | cap=6 | + R4 | ? |

238: 0.276 ? +R4 ****0.323****. Default-OFF; promote via config once R1's net-over-seg guard + corpus growth land.

247: milestone knobs OFF ? guards shipped behaviour) and ****MILESTONE**** (R3 cap=6 + R4 inactivity trim ? guards the

249: MILESTONE tolF1 ****0.323**** / f1@0.5 0.440 (matches the R3/R4 row above ? cross-checks the harness).

323:frame-aligned, added to `human\_lovo\_manifest.json`). ****n=8** milestone numbers (LOVO, tIoU-F1):**** baseline 0.232 ? W1**

324:0.403 ? W1+W2 0.419 ? ****W1+W2+hardcap 0.437**** (merge_rate 0.634 ? 0.000); milestone ? baseline?W1+W2 = ****+0.187 F1@0.5****.

325:All flags ****default-OFF****; promotion to a shipped milestone is owner-gated and re-run at the ~16-clip corpus.

333:every piece ? `max\_window\_duration`. Still only cuts (never merges ? recall can't drop). Threaded through

351:tradeoff and the motivation for the ****boundary head / rally-state-aware split**** (M1/R1), the next lever).

353:`max\_window\_duration` should be a true cap ? NOT by a golden lift (there is none at n=6). Promote/flip with

[Showing results with pagination = limit: 40]

18. assistant (2026-06-19T15:47:26.957Z)

This is very informative. The QUALITY_ITERATIONS.md R1 entry (line 104-107) confirms the cap=6 + R4 milestone is ****shipped**** (PR #204), flipped default-ON. Note line 219 says "R4 ? rally-END inactivity trim... ****default-OFF****" but that's in an older R1 entry (2026-06-17) that's now superseded by line 104's launched entry ? but QUALITY_ITERATIONS.md is NOT in my slice (my slice is CLAUDE.md, docs/README.md, docs/NEXT_STEPS.md).

Now, the key question for my slice: does CLAUDE.md, README.md, or NEXT_STEPS.md make a stale claim that local windowing is not-started / default-OFF / scaffolding-only?

CLAUDE.md line 16-21: the "Rally-detection quality track" is described as "***design merged 2026-06-13, owner-gated, NOT started***". But the cap=6+R4 windowing milestone HAS shipped (PR #204, default-ON in models.py). Let me check this carefully ? the CLAUDE.md line says the track is "NOT started" and Gate A is "not wired into production". Let me verify whether the windowing W1/R4 work is part of this "quality track" and whether the "NOT started" claim is now contradicted by shipped code.

Let me read the full CLAUDE.md quality-track bullet and check the R1 Launch Report and the served Gate A regression test.

19. user (2026-06-19T15:47:28.850Z)

```
104     ### R1 ? MILESTONE LAUNCHED (cap=6 + R4 flipped default-ON) ? re-verified at n=13
*(2026-06-18)*
105     The R1 milestone is **shipped** (PR #204): the W1 cap=6 + R4 inactivity-end defaults are
now ON.
106     Full quality-impact table on both rulers, the over-seg guard, and the honest caveats are
recorded in
107     the **[R1 Milestone Launch Report](R1_MILESTONE_LAUNCH_REPORT.md)** (per the Launch-
Report convention).
108
109     **Re-verified on the grown n=13 corpus** (the 2 new Boxhill clips GX010137/GX010141) ?
the milestone
110     holds and the headline is *stronger* than at n=11:
111
112     | transition | n=11 (filed) | **n=13 (launched)** |
113     |---|---|---|
114     | baseline ? milestone tolF1 | ?+0.232, 10/0, p=0.002 | **?+0.222, 12/0/1, p=0.0005** |
115     | cap12 ? cap6 tolF1 | ?+0.108, 10/0, p=0.002 | **?+0.110, 12/0/1, p=0.0005** |
116     | cap6 ? +R4 tolF1 | ?+0.047, 8/0, p=0.008 | ?+0.040, **9/1/3**, p=0.0215 |
117     | net over-seg guard | PASS (mr 0.694?0.150) | **PASS** (mr 0.651?0.161, 0 regressions)
|
118
119     Both new clips improve (GX010137 0.448?0.685, GX010141 0.333?0.431); tIoU ladder
baseline 0.236 ?
120     milestone 0.474. Independently reproduced by a 2nd harness (#214 `lovo_resweep.py`,
matches to 3 dp;
121     LOVO picks cap=6 **13/13** on tIoU) and survived a 5-lens adversarial workflow (every
LOO fold clears
122     the bar; worst-fold drop GX010128 still ?+0.202, p=0.0010 ? no single clip drives it).
**Honest notes:**
123     W1 cap is the dominant lever / R4 is the marginal +0.040 increment (1 loss = GX010137, a
tolerance
124     artifact); the gate is the NET rate-based guard (strict=False by design); GX010141 is
the worst-behaved
125     clip (future over-merge target). The nightly self-flags STALE post-flip ? re-baseline
once with
126     `--update-baseline`. W2 / hard-cap / R5 stay default-OFF.
127
128     ### R5 ? blob-fallback for the continuously-active blob (default-OFF, stratum tool)
*(2026-06-17)*
129     The one clip the cap6+R4 milestone still fails is the **continuously-active blob**
(mahadevpura_1:
130     ~174 s of play with no inactivity gap to split on). The W1 gap-split finds no gap; R4
finds no slow
131     tail; the result is one ~66 s mega-window that matches nothing (**tolF1 0.000**, the
degenerate
132     0-match outcome [`REAL_FOOTAGE_VALIDATION_REPORT.md`](REAL_FOOTAGE_VALIDATION_REPORT.md)
$5 R5 names).
133
134     **R5** (`window_blob_fallback`, default-OFF) is the last resort: **after** the gap-split
and the R4
```

135 trim have each had their chance, any group still longer than `window\_blob\_factor` × the
cap is a
136 genuine blob ? midpoint-chop it to ? the cap, **log the forced cut**, and **flag every
resulting
137 window `forced\_cut=True`** (surfacing the clip as one that needs rally-STATE cues ? net-
crossings /
138 two-sided motion ? not a bigger cap). `\_blob\_fallback\_chop` in `tracknet\_runner`, wired
through

20. user (2026-06-19T15:47:30.208Z)

ff51b37 chore(eval): re-baseline the nightly after the R1 milestone flip (#204) (#216)
67b6927 feat(config): R1 milestone ? flip windowing defaults ON (cap=6 + R4), re-verified at
n=13 (#204)
f5f339a fix(eval): fusion loaders skip windowing-only files on a mixed golden corpus (#210)
55e4c89 docs(archive): LOVO threshold re-sweep at n=13 ? milestone thresholds hold (#214)
8c68143 feat(eval): R1 ? net over-seg promotion guard unblocks the cap6+R4 milestone (#201)
bb4d6e2 feat(eval): Tier 1 (W1.5) ? hard-cap fallback enforces the window cap on continuously-
active trajectories (#195)
a3b209a feat(windowing): Tier 1 (W1) ? bounded windowing fixes the over-merge (?F1 +0.139, 6/0)
(#187)
ebdf331 fix: CI for PR171 - wrap long docstring lines (E501), use cast for json.load mypy
compat, ensure files end with newline
23ad934 fix: CI for PR170 - wrap long docstring line (ruff E501) in gate_verdict
b649f1c address review on #169: revert frozen archive edit (should-fix); trim handoff to short
pointer per #165 intent (record moved to plan); add Transition Record section; unbundle catalog
items 3/6/10 + clarify P3 'shipped' state with investigation (no fusion_compare.py etc. present;
general harness + P2 are); add M0 Quick Wins (the 4 in-container items per course-correction);
re-sequence milestones (M0 quick wins first + owner-led parallel, heavy last); update risk table
+ sequencing text. Thorough point-by-point in PR comment.
1b2450a docs: finalize handoff/audit after #166 merge + introduce post-166 work items plan with
risk assessment, milestones and sequencing (review first per owner)
4251c9c fix(eval): close the eval?serve windowing gap ? single-source SERVED preset + served-
regression promotion guard (#154)
53bc743 docs(quality): served-side Gate-A check ? flip's offline win doesn't transfer to
production windowing (#152)
7b9dad2 fix(tests): cross-platform support for Windows-style paths in eval batch resolver and
tracknet/wasb basename logic
8f4c564 Add real-time progress logging for MD5 hashing, add processing milestones tracking in
Gemini segmenter, and bypass heavy OpenCV validations on large files

21. assistant (2026-06-19T15:47:40.639Z)

Confirmed: PR #204 (commit 67b6927) "feat(config): R1 milestone ? flip windowing defaults ON
(cap=6 + R4)". The local windowing milestone is shipped/default-ON.

Now I need to assess my slice carefully:

1. **CLAUDE.md line 16**: "Rally-detection quality track (design merged 2026-06-13, owner-gated,
NOT started)". This is now contradicted ? the W1/R4 windowing milestone (part of this
quality track, RALLY_QUALITY_RESEARCH W1/R-series) HAS started AND shipped (PR #204 default-ON).
However, I need to be careful: this CLAUDE.md bullet specifically frames the track as the
fusion/Gate-A/experiment-harness sequence. The bullet also says Gate A is "not wired into
production." Let me check the precise scope. The phrase "NOT started" is a status claim that
contradicts shipped code. This is a factual staleness.

2. **CLAUDE.md line 61-62**: "This dev container lacks GPU / torch / cv2 / numpy ? the full
pipeline can't run here." SEED 1 ? confirmed stale (torch+cuda, cv2, numpy all present).

3. **NEXT_STEPS.md line 18**: "the dev box is CPU-only: `torch+cpu`/`cv2` present but no GPU" ?
stale (GPU present, torch+cu124).

4. **NEXT_STEPS.md line 69**: "Dev container no GPU/torch/cv2" ? stale.

Let me verify whether the CLAUDE.md "NOT started" is truly about the windowing track or strictly

about fusion/Gate-A. The bullet enumerates: HOW_RALLY_DETECTION_WORKS ? MULTI_SIGNAL_FUSION_PLAN (Gate A) ? EXPERIMENT_HARNESS ? R00RA. This is the fusion track specifically. The windowing (W1/R4) comes from RALLY_QUALITY_RESEARCH.md, which isn't in this bullet's list. So "NOT started" here is scoped to the fusion track. Let me check whether the fusion track has started (FusionSegmenter built, experiment harness built).

22. user (2026-06-19T15:47:41.298Z)

```
backend\pipeline\detectors\fusion_features.py
backend\eval\fusion_golden.py
backend\pipeline\segmenters\fusion_hybrid.py
backend\eval\fusion_audio.py
backend\eval\fusion_compare.py
```

23. user (2026-06-19T15:47:41.766Z)

```
backend\eval\experiment.py
```

24. user (2026-06-19T15:47:43.535Z)

```
__init__.py
__pycache__
base.py
fusion_hybrid.py
gemini.py
motion.py
tracknet_hybrid.py
trajectory_hybrid.py
wasb_hybrid.py
yolo_hybrid.py
---detectors---
__init__.py
__pycache__
audio_features.py
base.py
fusion_features.py
person_detector.py
rally_gate.py
tracknet_runner.py
wasb_infer.py
wasb_runner.py
```

25. assistant (2026-06-19T15:47:51.678Z)

The fusion track HAS started and shipped substantial code: `fusion\_hybrid.py` segmenter, `rally\_gate.py` (Gate A), `person\_detector.py`, `experiment.py`, `fusion\_compare.py`, `ablation.py`, `FusionConfig` in models.py. So CLAUDE.md line 16's "NOT started" for the rally-detection quality track is contradicted by code.

However ? I must be careful and surgical here. The CLAUDE.md "Current state" bullet (line 26-34) and the quality-track bullet (16-21) carry roadmap intent. The task says "Only fix factual contradictions, not roadmap intent." The "(design merged 2026-06-13, owner-gated, NOT started)" is a factual STATUS claim. Let me verify the served Gate A "not wired into production" claim precisely ? is Gate A now wired into the served path?

The FusionConfig docstring (models.py lines 84-114) says the fusion segmenter "does NOT gate the served handoff unless a threshold is raised (`min\_crossings`/`min\_players` default 0 = OFF)". So served Gate A is BUILT but default-OFF. The CLAUDE.md says Gate A is "***not wired into production**" ? but it IS now wired (available via `fusion\_hybrid` + `min\_crossings` config knob), just default-OFF. Let me check the served Gate A regression test referenced in NEXT_STEPS to confirm the precise state.

26. user (2026-06-19T15:47:54.127Z)

950106b fix(fusion): W2 net-crossings gate is do-no-harm ? never empty a non-empty video (#188 follow-up) (#190)
5cc0ea4 test + docs: hardening loop completion record + rally quality track P3 first step (harness LOVO for person features); identify next candidates (per NEXT_STEPS). Small landing comment in fusion_hybrid + explicit list in NEXT_STEPS. Syntax green. Pre: container baseline (ast + obsreport note). Branched, one item, will babysit. Verification: owner re-run full + real video.
cbf2236 feat(segmenters): FusionSegmenter ? shuttle + net-crossing + person-cue fusion (P2, default-OFF) (#117)
8615534 refactor: adopt approved layered directory structure
---when added---
cbf2236 feat(segmenters): FusionSegmenter ? shuttle + net-crossing + person-cue fusion (P2, default-OFF) (#117)

27. user (2026-06-19T15:47:55.679Z)

```
16:- ``_select_for_handoff`` ? optional served Gate A/B: when ``indexing.fusion.min_crossings``
39:class FusionSegmenter(TrajectoryHybridSegmenter):
84:                                     min_crossings=0,
estimate=self._axis_estimate(points))
150:     where no segment clears ``min_crossings``, the gate falls back to the full ungated
158:     min_crossings = int(fcfg.get("min_crossings", 0))
164:     if min_crossings > 0 and st.get("net_crossings", 0.0) < min_crossings:
176:segmenter_registry.register(FusionSegmenter)
```

28. assistant (2026-06-19T15:48:10.343Z)

Confirmed. The `FusionSegmenter` is registered (`fusion\_hybrid`), Gate A is available via the served path (`\_select\_for\_handoff` applies it when `min\_crossings > 0`), but it's default-OFF. So Gate A IS now "wired" into the segmenter, but default-OFF (not gating in production).

Now, the precise CLAUDE.md claim on line 19 reads: the held-shuttle bug is the already-built `rally\_gate.py` "Gate A" ****not wired into production****. At the time CLAUDE.md was written, Gate A was only called from eval drivers. Now (PR #117) it's wired into the `fusion\_hybrid` served segmenter via the `min\_crossings` knob ? but that segmenter is not the default (`default\_segmenter = "yolo\_hybrid"`) and the knob defaults to 0=OFF. So "not wired into production" is now partially stale: the wiring EXISTS (built, opt-in, default-OFF), it just isn't enabled by default.

This is nuanced. The seed for my slice focuses on the windowing milestone, and explicitly says "Only fix factual contradictions, not roadmap intent." The "Gate A not wired into production" claim is borderline ? the wiring code now exists but is default-OFF/not-the-default-segmenter, so "not wired into production" is arguably still accurate in the production-default sense. I'll be conservative and NOT flag the Gate A wiring line, since the default segmenter is still `yolo\_hybrid` and Gate A is default-OFF ? calling it "wired into production" would over-claim, and "not wired into production (by default)" is defensible. The strongest, cleanest contradiction is "NOT started" for the whole quality track.

Let me re-read CLAUDE.md line 16 precisely. It says the track's design "merged 2026-06-13, owner-gated, NOT started." But fusion_hybrid (#117), rally_gate wiring, experiment harness, AND the windowing milestone (#204, default-ON) have all shipped. "NOT started" is factually false. The minimal surgical fix: the track HAS started (multiple PRs shipped, windowing milestone default-ON). But I must preserve voice and not relitigate owner-gating intent.

Let me check the CLAUDE.md "Current state" para once more and decide. I'll also double check the NEXT_STEPS "implementation in progress" vs other "NOT started" language to avoid double-flagging where the doc is already correct. NEXT_STEPS line 54 already says "implementation in progress" ? so NEXT_STEPS is internally inconsistent but at least line 54 is current. The stale items in NEXT_STEPS are the GPU/CPU environment claims.

Let me verify the exact uniqueness of the strings I'll target, and confirm there's not another "lacks GPU" type claim I'm missing in README.

29. user (2026-06-19T15:48:12.236Z)

11:[Omitted long matching line]
14:[Omitted long matching line]
20:- GOLDEN_DATA_SHARING.md ? How the GPU box and CPU dev/agent sessions share golden artifacts (trajectory CSVs, `\*\_golden\_features.json`) via a OneDrive folder; the GPU-extract ? CPU-litmus split. See ****#175****.

30. user (2026-06-19T15:48:13.368Z)

8:[Omitted long matching line]
11:[Omitted long matching line]

31. assistant (2026-06-19T15:48:20.215Z)

README line 20's "GPU box and CPU dev/agent sessions" / "GPU-extract ? CPU-litmus split" describes the GOLDEN_DATA_SHARING workflow convention (a documented two-machine split convention), not a hard claim that THIS environment can't run GPU. That's a convention/workflow description, and it's tied to where the corpus/videos live (owner-side), not purely GPU availability. I'll be conservative and not flag README line 20 ? it describes an artifact-sharing convention rather than asserting "no GPU here." The README "design (owner-gated, NOT started)" tags on lines 8/11/14 are for individual design docs (ANALYZERS_AND_RECONCILERS, GOLDEN_REGRESSION_FIXTURES, DESKTOP_APP_PLAN) ? let me verify those are genuinely not-built before deciding, since a "[design] NOT started" tag is stale only if the thing is actually built.

32. user (2026-06-19T15:48:21.130Z)

8 - ANALYZERS_AND_RECONCILERS.md ? Design (owner-gated, NOT started): a ****signal-fusion framework**** for rally detection ? producers emit confidence-scored ****signals**** at three temporal ****scopes**** (frame detectors / window analyzers / session analyzers), a learned ****scorer**** weights them, and an auditable ****report-first reconciliation**** pass lints the decision against the evidence. The spine: ****no special-casing in the decision path**** (signals?scorer, never `if/else`). Worked examples: a service-fault analyzer, a warm-up/practice span detector, and consuming a per-frame shuttle-state detector. A peer to `EXPERIMENT\_HARNESS.md` and `MULTI\_SIGNAL\_FUSION\_PLAN.md` (ADR-007 modular rally layer).

9 - PERSONALIZATION_PLAN.md ? Design (owner-adopted ****hybrid****; Phase-0 landing) for the ****per-user personalization feature on a generalization-first baseline****: a thin per-user ****tuning* layer gated by the honest small-N statistics (a ****min-N promotion floor**** + a ****structural-guard exemption****, never a service gate) + a ****contribution flywheel**** (free-tier videos pool into the owner-trained baseline; ****"a tool that gets better the more you help it"**)**. Records the ****locked owner decisions**** (identity, storage, `min\_n` semantics, Gate-A default, free-tier pooling). Built so far: per-user promotion gate (#141) + held-out-USER learning curve (#142).

10

11 - GOLDEN_REGRESSION_FIXTURES.md ? ****Design (owner-gated, NOT started):**** video-free, non-attributable regression fixtures so a clean clone runs the rally-seg suite with ****no video/GPU/DB****. Two layers ? the post-detector freeze + ****dual gates**** (characterization + quality-floor) reusing the `backend/eval/` stack, and the ****SEALED-FIXTURES**** privacy filter (de-attribution-at-source, tiered public/key-gated, shuttle-only public tier). Carries a cited ****IN/US/EU legal memo**** (derived-data IP/privacy), a threat model, and a ****\$7 progress tracker + definition-of-done****. Extends GOLDEN_DATA_SHARING.md.

12

13 **## Product / platform plans**

14 - DESKTOP_APP_PLAN.md ? ****Tracked scoping doc (2026-06-18, owner-gated, `[design]` ? NOTHING built):**** package the local pipeline as a ****cross-platform desktop app**** for non-technical players ? plug in the camera's USB/SD card, click once, get back ****only the highlight reels****, fully on-device (no upload, no original-video bloat). The centerpiece is the ****de-WSL native-compute port**** (run WASB natively per-OS: Linux/Windows CUDA · macOS MPS/CPU ? drop WSL2), and its ****make-or-break unknown****: is CPU acceptable on a typical ****no-GPU* enthusiast laptop (WASB is ~3.4x real-time on a laptop 2070S)? \$7 tracker = ****P0 compute-feasibility spike ? P1 native detector ? P2 app shell ? P3 packaging/installers******. Realizes the ****L0 fully-local**** tier of VIDEO_LOCALITY_MODEL.md; is the ****`LocalDesktop` ComputeBackend**** of PLATFORM_ARCHITECTURE.md §3.2; the smaller/export-clean owned model

(OWNED_MODEL_IMPLEMENTATION_PLAN.md) is one answer to the compute crux.

15

33. user (2026-06-19T15:48:21.295Z)

No files found

34. assistant (2026-06-19T15:48:31.842Z)

ANALYZERS_AND_RECONCILERS (no analyzer/reconciler .py exists) and GOLDEN_REGRESSION_FIXTURES (a fixtures design) and DESKTOP_APP_PLAN are genuinely not-built design docs ? their "NOT started" tags are accurate, not stale. Good, I won't flag those.

Now let me make my final determination on the CLAUDE.md "Rally-detection quality track ... NOT started" claim. The fusion track (fusion_hybrid #117, rally_gate wiring, experiment.py harness) AND the windowing milestone (#204 default-ON) have shipped. So "NOT started" is factually contradicted. But I need to be precise about minimal surgical editing and preserving voice/owner-gating intent. The cleanest correction: the track is owner-gated but implementation has started (multiple PRs shipped, windowing milestone default-ON) ? matching NEXT_STEPS.md line 54 which already says "implementation in progress." Let me make the CLAUDE.md status consistent with that.

Let me also reconsider the "Gate A not wired into production" sub-claim within the same CLAUDE.md bullet. Since I'm editing that bullet anyway, but the rule is minimal surgical edits with unique stale_text ? I'll keep the two as separate, narrowly-scoped findings. Actually, for the CLAUDE.md bullet, the most defensible single contradiction is the "NOT started" parenthetical. The "Gate A not wired into production" is genuinely nuanced (default segmenter is still yolo_hybrid, Gate A default-OFF) ? I'll leave that phrase alone to avoid over-claiming. I'll narrowly fix only "NOT started."

Let me verify the exact text uniqueness for each finding before writing them.

35. user (2026-06-19T15:48:37.743Z)

```
=== CLAUDE.md NOT started count ===
1
=== CLAUDE.md lacks GPU count ===
1
=== NEXT_STEPS CPU-only count ===
1
=== NEXT_STEPS no GPU/torch/cv2 count ===
1
```

36. assistant (2026-06-19T15:48:47.661Z)

All four target strings are unique. Now let me finalize my findings. Let me reconsider the NEXT_STEPS.md line 18 carefully ? it says "the dev box is CPU-only: `torch+cpu`/`cv2` present but no GPU, and the corpus/videos/CSVs/labels are not here." The corpus-not-here part may still be true, but "CPU-only / torch+cpu / no GPU" is contradicted (torch 2.6.0+cu124, cuda True). The minimal surgical fix corrects only the GPU/torch claim while preserving the corpus-location claim.

Similarly NEXT_STEPS line 69: "Dev container no GPU/torch/cv2 ? Gate A, person cue logic, harness, and pure fusion paths are testable here; real-video / golden LOVO / ablation confirmation is owner-side (GPU/WSL machine)." The "no GPU/torch/cv2" is stale; the rest (corpus/video/golden owner-side) is a separate (still-plausible) claim.

Let me now write the structured output with my verified findings. I have four high-confidence factual contradictions:

1. CLAUDE.md line 61-62 ? "lacks GPU / torch / cv2 / numpy" (SEED 1)
2. NEXT_STEPS.md line 18 ? "the dev box is CPU-only: torch+cpu/cv2 present but no GPU"
3. NEXT_STEPS.md line 69 ? "Dev container no GPU/torch/cv2"

4. CLAUDE.md line 16 ? quality track "NOT started" (contradicted by shipped fusion track + #204 windowing default-ON)

For #4, I'll keep the edit minimal: change "NOT started" to reflect that implementation has started (consistent with NEXT_STEPS line 54 "implementation in progress"), preserving the "owner-gated" framing (roadmap intent untouched).

37. user (2026-06-19T15:49:02.775Z)

Structured output provided successfully

38. assistant (2026-06-19T15:49:15.781Z)

I've completed the staleness audit of my slice (CLAUDE.md, docs/README.md, docs/NEXT_STEPS.md) and returned four verified findings via StructuredOutput.

Summary of what I verified and found:

****Confirmed stale (4 findings):****

1. ****CLAUDE.md:61-62**** ? "This dev container lacks GPU / torch / cv2 / numpy ? the full pipeline can't run here." Directly contradicted: ``torch 2.6.0+cu124`` with ``cuda.is_available()==True``, ``cv2 4.13.0``, ``numpy 2.4.6`` all import here. (SEED 1 confirmed.)

2. ****docs/NEXT_STEPS.md:18**** ? "the dev box is CPU-only: ``torch+cpu``/``cv2`` present but no GPU." Same contradiction; corrected while preserving the still-true "corpus/videos not here" part.

3. ****docs/NEXT_STEPS.md:69**** ? "Dev container no GPU/torch/cv2." Same contradiction.

4. ****CLAUDE.md:16**** ? Rally-detection quality track tagged "NOT started." Contradicted by shipped code: ``fusion_hybrid.py`` (#117, registered segmenter), ``rally_gate.py`` Gate A wired into the served handoff, ``backend/eval/experiment.py`` harness, and the W1 cap=6 + R4 windowing milestone flipped ****default-ON**** in PR #204 (``backend/config/models.py``: ``max_window_duration=6.0``, ``window_inactivity_end=1.5``, ``inactivity_rally_thresh=0.20``, ``inactivity_min_density=0.30``). NEXT_STEPS.md:54 already says "implementation in progress" for the same track. (SEED 2 corroborated.)

****Checked and deliberately NOT flagged (not stale / out of scope):****

- CLAUDE.md "Gate A not wired into production" (line 19) ? wiring now exists via ``fusion_hybrid`` + ``min_crossings``, but default segmenter is still ``yolo_hybrid`` and the knob defaults to 0=OFF, so "not wired into production (by default)" remains defensible; calling it production-wired would over-claim.

- CLAUDE.md "Current state / Scaffolding complete / No platform/owned-model implementation has started" ? owner-gated roadmap intent about the platform/owned-model tracks, not the local windowing track; out of scope per instructions.

- README.md "[design] NOT started" tags for ANALYZERS_AND_RECONCILERS (no analyzer/reconciler ``.py`` exists), GOLDEN_REGRESSION_FIXTURES, and DESKTOP_APP_PLAN ? genuinely unbuilt, tags accurate.

- README.md:20 GOLDEN_DATA_SHARING "GPU box ? CPU dev/agent" split ? describes an artifact-sharing convention tied to corpus location, not a hard "no GPU here" assertion.

The corrections preserve each doc's voice and owner-gating framing, touching only the factual GPU/environment and track-status contradictions.